

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

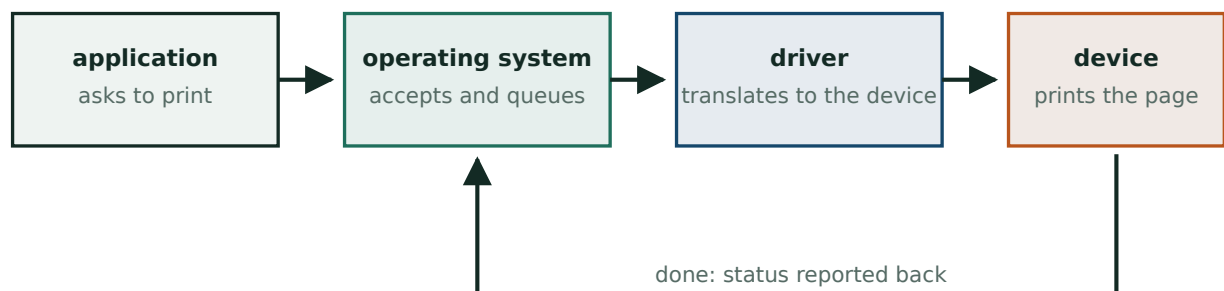
Topic 3.3 - Managing input and output

A real classroom situation

The teacher plugs a new printer into the lab's computer. Nothing happens for a few seconds; then a small message appears: "installing device software". A minute later the printer works, and - the remarkable part - the word processor, the browser and the results spreadsheet can all print to it, though none of them was written with this printer in mind. What happened in that minute is the quiet centre of this topic: the operating system installed a device driver, the translator that lets any application talk to any device, and took over the job of managing the traffic between them.

The journey of a print request

Four links between a click and a printed page



Buffers hold data in transit; spooling queues whole jobs so three teachers can click Print at once and the pages still come out in order.

Diagnostic ladder: device, driver, or application - name the rung before you shrug.

textbook.com.pk

Explanation

Managing input and output

Every device - keyboard, screen, disk, printer, USB drive - speaks its own low-level language of signals and registers. No application wants to know those languages, and no device maker wants to write a different version of every application. The operating system stands in the middle and makes both sides simple.

The chain works like this. An **application** asks for something ordinary: "print this page", "read this file", "show this frame". It addresses the request to the operating system, never to the device directly. The **operating system** accepts the request, checks that it is allowed, and places it in a queue. The **driver**, a small piece of system software written for one specific device, translates the request into that device's own commands. The **device** does the work and reports back, and the operating system passes the result to the application. Your click on "Print" and the whir of the printer are the two visible ends of that chain; the middle is the OS doing what it does all day.

Two more ideas complete the picture. A buffer is a small holding area in memory where the OS parks data in transit - the first part of a print job sits in a buffer so the application can move on while the slow printer catches up. And spooling is the same idea at larger scale: print jobs queue up one after another, which is why three teachers can click Print at once and the pages still come out in order, one job at a time.

Everyday requests and what the OS does

Request	What the OS does	What you notice
print a page	queues the job, translates via the driver, spools it to the printer	pages appear in order
open a file	finds the path in the file system, asks the disk driver to read	the document appears
save to a USB drive	writes via the USB driver in small buffered pieces	the progress bar fills
type on the keyboard	the keyboard driver reports each key press	letters appear on screen
plug in a new device	loads and installs the matching driver	the installing message, then the device works

Every row follows the same chain: application, OS, driver, device.

textbook.com.pk

textbook.com.pk

The payoff of this arrangement is enormous and usually invisible: applications stay simple because they speak one language (the OS), and devices stay interchangeable because each plugs in through its own driver. When a device "does not work", the chain gives you a diagnostic ladder: is it the device (does it have power?), the driver (does the OS recognise it?), or the application (does the request even reach the OS)? Naming the rung turns a shrug into a plan.

A note on the blocked state from Topic 3.2: now you can see why a process that starts an input/output operation goes blocked. The disk will take its time whatever the process does, so the OS parks the process and gives the processor to someone else until the device reports completion. Waiting is not wasted; it is scheduled.

Activity: The diagnostic ladder

Work in pairs. About 20 minutes.

1. Your teacher describes five faults: a printer that shows no lights; a printer with lights that no program can print to; a USB drive that works on one

computer but not another; a keyboard that types nothing; a file that opens on the office machine but not on the lab machine.

2. For each fault, agree where the chain is most likely broken: device, driver, or application - and write one sentence naming the evidence that points there.
3. For one fault of your choice, write the two questions you would ask the user first.
4. Compare ladders with another pair. Where you disagree, test both against the chain in the first figure.

Check your understanding

1. Define a device driver in one sentence. (*Remember*)
2. Trace the journey of a print request from the word processor to the page, naming all four links in the chain. (*Understand*)
3. A USB drive works on one computer but not another. Using the chain, give the most likely broken link and your reason. (*Apply*)
4. Explain what a buffer does and why printing does not freeze the word processor. (*Understand*)

Summary

Input/output management is the operating system's translation service: applications make ordinary requests, the OS queues and polices them, drivers translate them into each device's own language, and buffers and spooling smooth the speed differences so everything appears to happen at once. The chain doubles as a diagnostic ladder - device, driver, or application - and it explains why a process doing input/output goes blocked rather than busy.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the four links of an input/output request and say what each does.
- ☐ I can define driver, buffer and spooling with an example of each.

- ☐ I can use the chain to place a described fault on the right rung.

Try this at your practicum school

Find one device that is currently not working in the school (or ask about the last one that failed). Write three sentences: the symptom, which link of the chain the fault belonged to, and what fixed it.

Summative task

A teacher complains: "I clicked Print, the printer has power and paper, but nothing comes out, and only my computer has this problem; others print fine." In one page, analyse the fault using the input/output chain: name the link you would check first and why, what the next two checks would be, and what the likely fix is at each step.

Marking guide (10 marks):

- The chain named and applied to the symptom (Analyse): 4 marks.
- Checks ordered sensibly with a reason for each: 3 marks.
- Likely fixes matched to the right link: 2 marks.
- Clear writing: 1 mark.

Further reading

- Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2023). *Operating Systems: Three Easy Pieces* (version 1.10), chapters 4 (the blocked state) and 36 (I/O devices). <https://pages.cs.wisc.edu/~remzi/OSTEP/>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 2. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Norton, P. *Introduction to Computers*. McGraw-Hill Education. (Course guide reading; print monograph, cited at bibliographic level.)

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

[Give feedback](#)